



1 - INTRODUCTION

Gelucire® is a family of semi-solid to solid polyoxyglycerides able to self-emulsify in contact with aqueous phases. The new Gelucire® 48/16 is a solid waxy material, characterized by a drop point of 48°C and a HLB of 16. It is composed of PEG-32 mono- and diesters of palmitostearic acids - hydrophilic surfactants -, and free PEG-32 - hydrosoluble cosolvent. Hence, it corresponds to a Type IV formulation from the Lipid Formulation Classification System [1]. The aim of this study is to characterize this new excipient in terms of its capacity to finely disperse in aqueous phases, its digestibility, and its ability to maintain poorly water-soluble drugs in solution after dispersion and digestion.

2 - EXPERIMENTAL METHODS

• Binary phase diagram

The evolution of the macroscopic appearance of Gelucire® 48/16-water mixtures was assessed at 45°C under moderate magnetic stirring (200 rpm). Microscopic evaluation was performed using the Axio Scope A1 (Carl Zeiss) at 45°C to avoid the crystallization of the excipient during the test. Particle size distributions were measured by photon correlation spectroscopy (PSS 380 ZLS Nicomp). Viscosities of mixtures were measured at 45°C with a Rheomat 115 (Contraves).

• Solubility study in solid excipient

The solubility of ibuprofen in Gelucire® 48/16 was evaluated by DSC (Pyris Diamond, Perkin Elmer). A range of Gelucire® 48/16-ibuprofen mixtures was prepared with API concentration ranging from 0 to 750 mg/g. Mixtures were heated at 50°C and let at room temperature for two days to equilibrate. Then, each sample was analyzed by DSC from -20 to 100°C at 3°C/min. The melting endotherm of Gelucire® was integrated and melting enthalpy calculated. This enthalpy was plotted as a function of the API concentration. Furthermore, a binary solution of Gelucire® 48/16 with ibuprofen was developed.

• Lipolysis study

pH-stat apparatus (Titrand 842, Metrohm) was used to assess excipient digestibility by pancreatin [2, 3]. Enzymatic solutions were freshly prepared each day using 0.2g of pancreatin per milliliter of lipolysis medium (NaCl 150 mM; NaTC 4 mM; CaCl₂ 1.4 mM; Tris-HCl 1 mM). This solution was centrifuged at 5°C, 1520 RCF for 10 minutes (Universal centrifuge 320R). The supernatant was sampled and used for lipolysis.

Lipolysis in fasted state: 28.5 mL of lipolysis medium pH 2.2 was introduced in the pH-STAT vessel and thermostated at 37°C. Then 1.5g of solid Gelucire® 48/16 was added to the medium. It was necessary to wait about 15 minutes to have a complete dissolution of Gelucire® 48/16 in the aqueous medium. At t=0, 6mL of lipolysis medium pH 2.2 was added to simulate the gastric phase (gastric secretion). Three samples of 1mL were taken and rapidly frozen in liquid nitrogen at time 0, 15, and 29 minutes. To simulate the duodenal phase 4mL of a pancreatin solution (pancreatic secretion) was added in the reaction vessel at time 30 minutes. The pH was shifted at 5.7. Five samples of 1mL were taken and rapidly frozen in liquid nitrogen at 35, 40, 45, 60, and 90 minutes. All samples were stored in the freezer until freeze-drying and subsequent HPLC-ELSD analysis.

Lipolysis in fed state: same conditions as above but the gastric and duodenal pHs were set at 5.7 and 6.25, respectively. The volume of pancreatic solution added at time 30 minutes was 22 mL to dilute the gastric by 1.7-fold [4].

The efficiency to maintain the drug soluble during digestion was assayed using the pH-stat apparatus [4]. The lipolysis conditions were the same as for the determination of the metabolites of Gelucire® 48/16 in the fasted or fed state, except for (i) the introduction of the sample in the vessel at t=0, and (ii) the sample preparation: after each sampling 5μL of 4-bromophenylboronic acid was quickly added to stop the hydrolysis of the excipient and then the sample was centrifuged at 37°C, 21380 RCF for 10 minutes. The micellar phase was then collected and diluted with acetonitrile (1:1) in HPLC vials. Ibuprofen concentration was measured by HPLC-DAD.

3 - RESULTS AND DISCUSSION

The addition of a few drops of water to Gelucire® 48/16 hydrates the PEG chains. The mixture remained a limpid liquid up to 7% of water. Further addition of water led to the formation of a gel. This gel phase was observed between 7 and 73% of water. The viscosity of this gel increases up to 50% water and reaches a maximum of 4 Pa.s (Figure 1). Further addition of water leads to a reduction of viscosity. Between 73 and 80% of water, the viscosity of the gel continues to decrease to form a liquid limpid gel and then after 81% of water a liquid limpid solution. This solution is a micellar solution with particle size of 4.2 and 1.7nm for 80 and 90% of water, respectively. These experiments showed that Gelucire® 48/16 is highly hydrophilic. Emulsification process starts with the formation of a gel after addition of small amount of water. The mixture is organized in a lamellar structure as observed by optical microscopy under polarized light. This hydration/gelation phase is followed by the erosion of the gel phase. Increasing amounts of water promote the transition from the lamellar phase to the micellar solution.

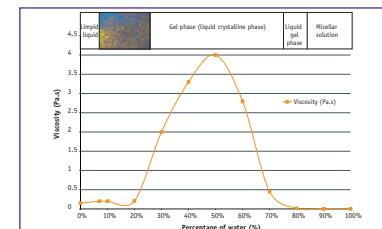


Figure 1: Binary phase diagram of Gelucire® 48/16 and water

Gelucire® 48/16 exhibits low digestibility by pancreatin in fasted conditions. The analysis of the composition shows a slow and steady hydrolysis of PEG-32 monoesters and diesters and the formation of free PEG (figure 2). After one hour of duodenal lipolysis about 50% of PEG-32 esters are still not degraded. PEG-32 monoesters show a more pronounced hydrolysis than PEG-32 diesters and are a better substrate for pancreatic enzymes, especially Carboxyl Ester Hydrolase (CEH) as demonstrated in [5].

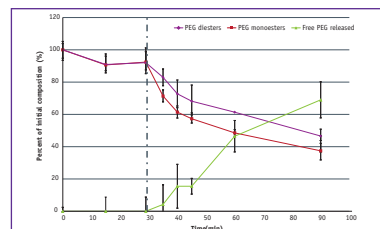


Figure 2: Evolution of the composition of each component of Gelucire® 48/16 during its in vitro lipolysis in fasted conditions. The vertical line at 30 min of elapsed time marks the end of the gastric phase and the beginning of the pancreatic phase.

Figure 3 shows that Gelucire® 48/16 is rapidly hydrolyzed in fed conditions. Indeed the quantity of pancreatic solution is increased by 5.5-fold and the pH is raised to 6.25, leading to conditions more favorable to lipid digestion. In fed conditions, all PEG-32 esters are hydrolyzed after only 30 minutes of duodenal phase. Accordingly the content of free PEG is dramatically increased.

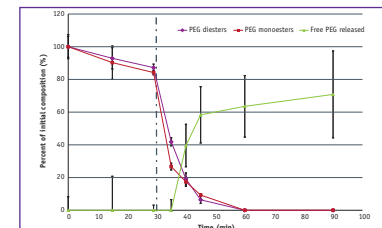


Figure 3: Evolution of the composition of each component of Gelucire® 48/16 during its in vitro lipolysis in fed conditions. The vertical line at 30 min of elapsed time marks the end of the gastric phase and the beginning of the pancreatic phase.

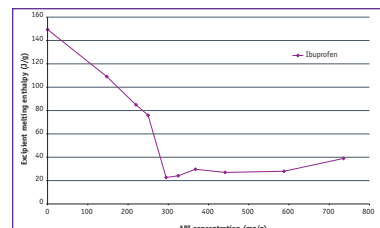


Figure 4: Evolution of the melting enthalpy of Gelucire® 48/16 as a function of ibuprofen concentration.

Figure 5 shows that the concentration of ibuprofen in the micellar phase increases during the first 15 minutes of the experiments due to the hydration / micellisation process of Gelucire® 48/16. Ibuprofen is more soluble in the simulated fed gastric phase than in the fasted one because it is weak acid. In fasted conditions, the concentration of ibuprofen in the 'duodenal' phase increases due to the pH shift to 5.7. No significant difference are observed in fasted conditions, the lipolysis of Gelucire® 48/16 in this condition has no effect on its ability to maintain in solution the drug. A slight supersaturation is even observed at the end of the experiment as the concentration in the micellar phase is above ibuprofen solubility in the digested milieu (118%). In fed conditions, the concentration of ibuprofen in the micellar phase decreases significantly due to the lipolysis of the excipient. However, the same level of supersaturation is observed as the concentration of ibuprofen at the end of the experiment is at 122% of its solubility. In addition, even if the excipient is completely hydrolyzed after 30 minutes of duodenal phase, the concentration of ibuprofen in the micellar phase is high and always above 60%, showing the solubilizing effect of free PEG after the hydrolysis of the micelles formed by PEG-32 esters.

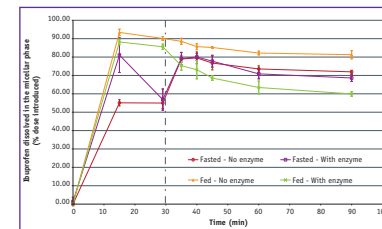


Figure 5: Variation in the concentration of ibuprofen dissolved in the aqueous phase, expressed in percent of the dose introduced, during the gastrointestinal lipolysis of a formulation containing Gelucire® 48/16 by pancreatin, in fasted or fed conditions, as a function of time. The vertical line at 30 min of elapsed time marks the end of the gastric phase and the beginning of the pancreatic phase.

4 - CONCLUSION

Gelucire® 48/16 self-emulsifies in contact with aqueous phases to form micellar solution. This excipient is a good solvent for poorly water-soluble drugs such as ibuprofen and is not prone to extensive digestion in fasted conditions. However, under fed conditions, the excipient is completely digested after 30 minutes of duodenal phase. The lipolysis of formulations containing ibuprofen as a solution show that Gelucire® 48/16 is able to maintain the drug in solution in fasted and fed conditions. This excipient is a promising new surfactant for self emulsifying systems.

5 - REFERENCES

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